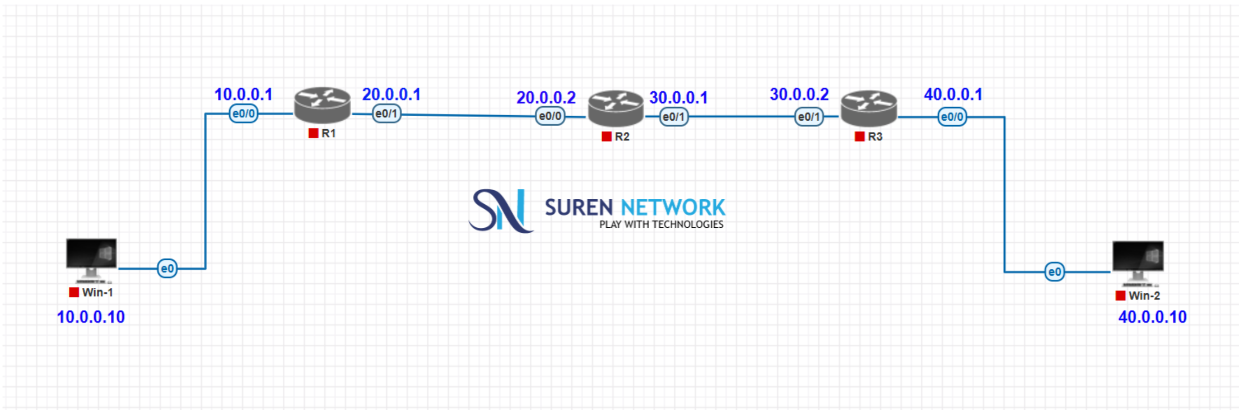
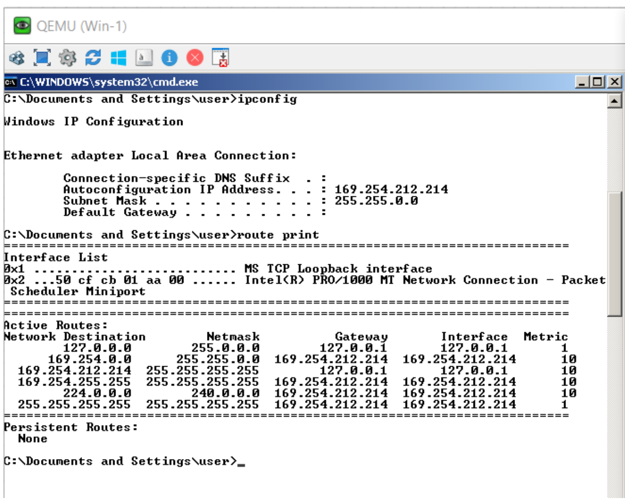
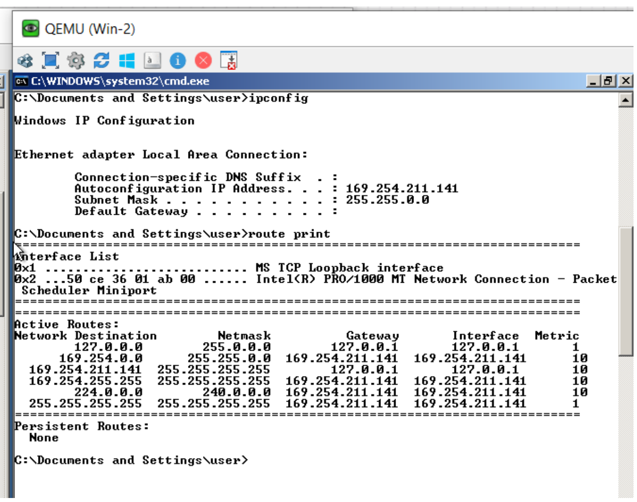
# Static Route

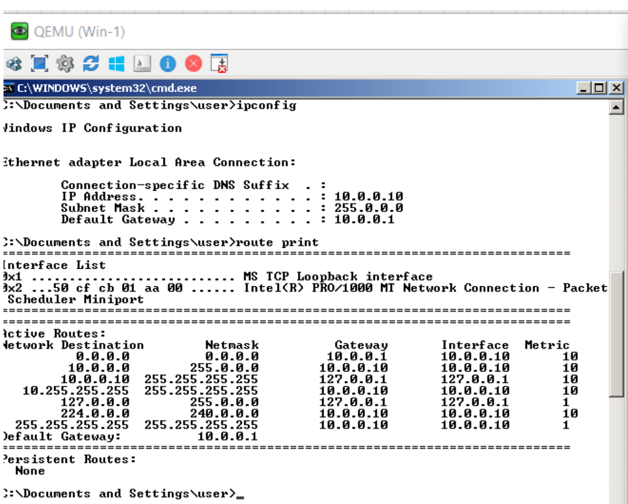


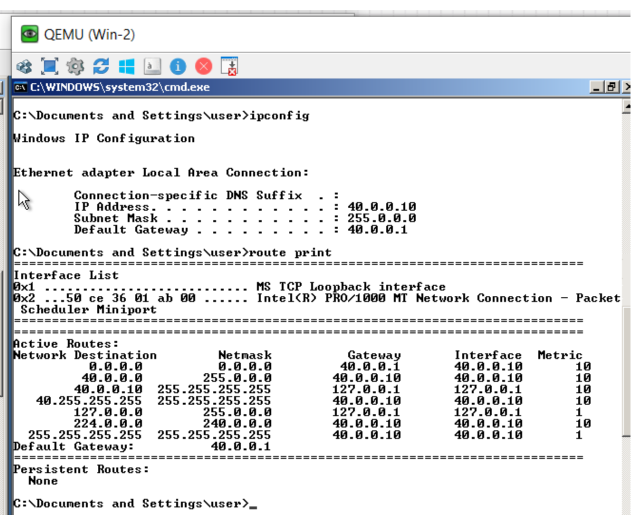

## Before the Ip configuration In PCs




## After the Ip configuration In PCs



Assign ip address in all cisco routers

R1

```
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#int e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#do sh ip add br
                       ^
% Invalid input detected at '^' marker.

R1(config)#do sh ip int  br
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                10.0.0.1        YES manual up                    up

Ethernet0/1                20.0.0.1        YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R1(config)#
```

R2

```
R2(config)#
R2(config)#int e0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exi
R2(config)#
R2(config)#int e0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
R2(config)#do sh ip int br
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                20.0.0.2        YES manual up                    up

Ethernet0/1                30.0.0.1        YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R2(config)#
```

R3

```
R3(config)#int e0/0
R3(config-if)#ip add 40.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#ex
R3(config)#
R3(config)#int e0/1
R3(config-if)#ip add 30.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exi
R3(config)#
R3(config)#do sh ip int br
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                40.0.0.1        YES manual up                    up

Ethernet0/1                30.0.0.2        YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down
```

R1

```
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/8 is directly connected, Ethernet0/0
L        10.0.0.1/32 is directly connected, Ethernet0/0
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/1
L        20.0.0.1/32 is directly connected, Ethernet0/1
R1(config)#
```

R2

```
Gateway of last resort is not set

      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/0
L        20.0.0.2/32 is directly connected, Ethernet0/0
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/1
L        30.0.0.1/32 is directly connected, Ethernet0/1
R2(config)#
```

R3

```
Gateway of last resort is not set

      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/1
L        30.0.0.2/32 is directly connected, Ethernet0/1
      40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        40.0.0.0/8 is directly connected, Ethernet0/0
L        40.0.0.1/32 is directly connected, Ethernet0/0
R3(config)#
```

Add static route for missing route in router-1

```
R1
R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.2
R1(config)#
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

Add static route for missing route in router-2

```
R2

Router(config)#ip ro
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
Router(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
Router(config)#
```

Add static route for missing route in router-3

```
R3
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#
R3(config)#
```

Check routing table for new routes in routing table

```
R1

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S    30.0.0.0/8 [1/0] via 20.0.0.2
S    40.0.0.0/8 [1/0] via 20.0.0.2
```

```
R2
S    10.0.0.0/8 [1/0] via 20.0.0.1
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.2/32 is directly connected, Ethernet0/0
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/1
L       30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
```

```
R3

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/1
L       30.0.0.2/32 is directly connected, Ethernet0/1
     40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.0.0.0/8 is directly connected, Ethernet0/0
L       40.0.0.1/32 is directly connected, Ethernet0/0
```

Ping 10.0.0.10 to 40.0.0.10